

Full Algebraic Derivation of 2D Radiation Diffusion Implicit Stencil

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1 Continuous Equations in Cylindrical Coordinates

We start with the coupled radiation and material energy equations in axisymmetric cylindrical coordinates (r, z, t) .

1.1 Radiation Energy Equation

The radiation energy density $E(r, z, t)$ evolves according to:

$$\frac{\partial E}{\partial t} = \frac{c}{3\sigma} \left(\frac{\partial^2 E}{\partial r^2} + \frac{1}{r} \frac{\partial E}{\partial r} + \frac{\partial^2 E}{\partial z^2} \right) - \chi c \sigma (E - U), \quad (1)$$

where

- c is the speed of light,
- $\sigma(T)$ is the temperature-dependent opacity (Rosseland mean), which varies in space via temperature,
- $U(r, z, t)$ is the material energy density (to be coupled below),
- the term $\frac{1}{r} \frac{\partial E}{\partial r}$ arises from cylindrical geometry.

Important: The form in Eq. (1) is written for notational simplicity. Since $\sigma(T(r, z))$ is *space-dependent*, the operator $\frac{c}{3\sigma} \nabla^2$ is not linear in the usual sense. In discretization, we must use the *conservative form* with diffusion coefficient $D(r, z) = \frac{c}{3\sigma(T(r, z))}$ varying at each location, giving:

$$\frac{\partial E}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D \frac{\partial E}{\partial r} \right) + \frac{\partial}{\partial z} \left(D \frac{\partial E}{\partial z} \right) - \chi c \sigma (E - U). \quad (2)$$

This is the form we use throughout the derivation below.

1.2 Material Energy Equation

The material energy density $U(r, z, t)$ evolves according to:

$$\frac{1}{\beta} \frac{\partial U}{\partial t} = \chi c \sigma (E - U), \quad (3)$$

where $\beta = \frac{\partial U_R}{\partial U_m}$ is the ratio of radiation to material heat capacity (dimensionless coupling strength).

1.3 Decomposition of the Diffusion Operator

Define the space-dependent diffusion coefficient:

$$D(r, z, t) \equiv \frac{c}{3\sigma(T(r, z, t))}. \quad (4)$$

The diffusion operator in conservative form (Eq. (2)) is:

$$\mathcal{L}[E] \equiv \frac{1}{r} \frac{\partial}{\partial r} \left(r D \frac{\partial E}{\partial r} \right) + \frac{\partial}{\partial z} \left(D \frac{\partial E}{\partial z} \right) = \mathcal{L}_r[E] + \mathcal{L}_z[E], \quad (5)$$

where the radial part is:

$$\mathcal{L}_r[E] \equiv \frac{1}{r} \frac{\partial}{\partial r} \left(r D \frac{\partial E}{\partial r} \right) \quad (6)$$

and the axial part is:

$$\mathcal{L}_z[E] \equiv \frac{\partial}{\partial z} \left(D \frac{\partial E}{\partial z} \right). \quad (7)$$

In the numerical implementation, we discretize using a finite-volume approach where $D(r, z)$ is evaluated at cell faces and treated as piecewise constant over each control volume. The diffusion coefficients $D_{i,j\pm 1/2}^n$ are then lagged in time (evaluated at t^n from U^n).

The coupled equations in conservative form are:

$$\frac{\partial E}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D \frac{\partial E}{\partial r} \right) + \frac{\partial}{\partial z} \left(D \frac{\partial E}{\partial z} \right) - \chi c \sigma (E - U), \quad (8)$$

$$\frac{1}{\beta} \frac{\partial U}{\partial t} = \chi c \sigma (E - U). \quad (9)$$

2 Time Discretization via Backward Euler

We discretize time using the backward (implicit) Euler scheme on a sequence of times

$$t^n = t^{n-1} + \Delta t,$$

where $\Delta t = \Delta t_n$ may vary with n .

2.1 Implicit Euler for Radiation Energy

Apply backward Euler to Eq. (8):

$$\frac{E^{n+1} - E^n}{\Delta t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D^n \frac{\partial E^{n+1}}{\partial r} \right) + \frac{\partial}{\partial z} \left(D^n \frac{\partial E^{n+1}}{\partial z} \right) - \chi c \sigma^n (E^{n+1} - U^{n+1}), \quad (10)$$

where we use time-lagged space-dependent coefficients $D^n(r, z)$ and $\sigma^n(r, z)$ (computed from U^n) at the old time, while spatial derivatives of E are evaluated at the new time (implicit in space, explicit in the coefficients).

Rearrange:

$$\frac{E^{n+1}}{\Delta t} - \frac{1}{r} \frac{\partial}{\partial r} \left(r D^n \frac{\partial E^{n+1}}{\partial r} \right) - \frac{\partial}{\partial z} \left(D^n \frac{\partial E^{n+1}}{\partial z} \right) + \chi c \sigma^n E^{n+1} - \chi c \sigma^n U^{n+1} = \frac{E^n}{\Delta t}. \quad (11)$$

2.2 Implicit Euler for Material Energy

Note on LTE coupling: The coupling factor χ (dimensionless) modulates the interaction strength between radiation and material in Local Thermodynamic Equilibrium (LTE). The factor appears in $A^n = \beta^n \Delta t \cdot \chi \cdot c\sigma^n$, with typical values $\chi = 1$ for full coupling or $\chi < 1$ for reduced coupling.

Apply backward Euler to Eq. (9), using time-lagged coefficients β^n and σ^n (computed from U^n):

$$\frac{1}{\beta^n} \frac{U^{n+1} - U^n}{\Delta t} = \chi c\sigma^n (E^{n+1} - U^{n+1}). \quad (12)$$

Rearrange:

$$\frac{1}{\beta^n} U^{n+1} + \Delta t \cdot \chi \cdot c\sigma^n U^{n+1} = \frac{1}{\beta^n} U^n + \Delta t \cdot \chi \cdot c\sigma^n E^{n+1}. \quad (13)$$

Factor:

$$\left(\frac{1}{\beta^n} + \Delta t \cdot \chi \cdot c\sigma^n \right) U^{n+1} = \frac{1}{\beta^n} U^n + \Delta t \cdot \chi \cdot c\sigma^n E^{n+1}. \quad (14)$$

Define:

$$A^n \equiv \beta^n \Delta t \cdot \chi \cdot c\sigma^n. \quad (15)$$

Then:

$$(1 + A^n) U^{n+1} = U^n + A^n E^{n+1}. \quad (16)$$

Solve for U^{n+1} in terms of E^{n+1} :

$$U^{n+1} = \frac{U^n + A^n E^{n+1}}{1 + A^n} \quad (17)$$

2.3 Substituting the Material Update into the Radiation Equation

Substitute Eq. (17) into the radiation equation:

$$\frac{E^{n+1}}{\Delta t} - \frac{1}{r} \frac{\partial}{\partial r} \left(r D^n \frac{\partial E^{n+1}}{\partial r} \right) - \frac{\partial}{\partial z} \left(D^n \frac{\partial E^{n+1}}{\partial z} \right) + \chi c\sigma^n E^{n+1} - \chi c\sigma^n \frac{U^n + A^n E^{n+1}}{1 + A^n} = \frac{E^n}{\Delta t}. \quad (18)$$

Expand the coupling term:

$$\chi c\sigma^n \frac{U^n + A^n E^{n+1}}{1 + A^n} = \frac{\chi c\sigma^n U^n}{1 + A^n} + \frac{\chi c\sigma^n A^n E^{n+1}}{1 + A^n}. \quad (19)$$

Substitute back:

$$\frac{E^{n+1}}{\Delta t} - \frac{1}{r} \frac{\partial}{\partial r} \left(r D^n \frac{\partial E^{n+1}}{\partial r} \right) - \frac{\partial}{\partial z} \left(D^n \frac{\partial E^{n+1}}{\partial z} \right) + \chi c\sigma^n E^{n+1} - \frac{\chi c\sigma^n A^n E^{n+1}}{1 + A^n} = \frac{E^n}{\Delta t} + \frac{\chi c\sigma^n U^n}{1 + A^n}. \quad (20)$$

Combine the E^{n+1} coefficients on the left:

$$\frac{1}{\Delta t} + \chi c\sigma^n - \frac{\chi c\sigma^n A^n}{1 + A^n}. \quad (21)$$

Simplify the coupling diagonal:

$$\chi c \sigma^n - \frac{\chi c \sigma^n A^n}{1 + A^n} = \chi c \sigma^n \left(1 - \frac{A^n}{1 + A^n} \right) \quad (22)$$

$$= \chi c \sigma^n \left(\frac{1 + A^n - A^n}{1 + A^n} \right) \quad (23)$$

$$= \frac{\chi c \sigma^n}{1 + A^n}. \quad (24)$$

So the equation becomes:

$$\left(\frac{1}{\Delta t} + \frac{\chi c \sigma^n}{1 + A^n} \right) E^{n+1} - \frac{1}{r} \frac{\partial}{\partial r} \left(r D^n \frac{\partial E^{n+1}}{\partial r} \right) - \frac{\partial}{\partial z} \left(D^n \frac{\partial E^{n+1}}{\partial z} \right) = \frac{E^n}{\Delta t} + \frac{\chi c \sigma^n}{1 + A^n} U^n. \quad (25)$$

Define the effective coupling coefficient:

$$C^n \equiv \frac{\chi c \sigma^n}{1 + A^n}. \quad (26)$$

Then Eq. (25) reads:

$$\left(\frac{1}{\Delta t} + C^n \right) E^{n+1} - \frac{1}{r} \frac{\partial}{\partial r} \left(r D^n \frac{\partial E^{n+1}}{\partial r} \right) - \frac{\partial}{\partial z} \left(D^n \frac{\partial E^{n+1}}{\partial z} \right) = \frac{E^n}{\Delta t} + C^n U^n. \quad (27)$$

3 Spatial Discretization

3.1 Grid Definition

We discretize the domain into a 2D grid of cell-centers:

$$z_i = (i - 1) \Delta z, \quad i = 0, 1, \dots, N_z - 1, \quad \Delta z = \frac{L_z}{N_z - 1}, \quad (28)$$

$$r_j = \text{general radial nodes}, \quad j = 0, 1, \dots, N_r - 1. \quad (29)$$

In general, r_j may be non-uniform (e.g., uniform in the foam region and geometrically stretched in the gold layer). For simplicity in the derivation, we first assume uniform Δr , then note the modifications for non-uniform grids.

Stored fields are cell-centered:

$$E_{i,j}^n \approx E(r_j, z_i, t^n), \quad U_{i,j}^n \approx U(r_j, z_i, t^n). \quad (30)$$

3.2 Face Values and Averaging

Spatial derivatives are computed at cell faces. The faces are located at:

$$r_{j-\frac{1}{2}} = \frac{r_{j-1} + r_j}{2}, \quad \Delta r_j = r_j - r_{j-1}, \quad (31)$$

$$z_{i-\frac{1}{2}} = \frac{z_{i-1} + z_i}{2}, \quad \Delta z = z_i - z_{i-1}. \quad (32)$$

For space-dependent coefficients like $D(T(r, z))$, we evaluate T at cell centers to get $D_{i,j}$, then average D to each face. Common averaging methods are:

$$\text{Arithmetic: } D_{i,j+\frac{1}{2}} \approx \frac{D_{i,j} + D_{i,j+1}}{2}, \quad (33)$$

$$\text{Harmonic: } D_{i,j+\frac{1}{2}} \approx \frac{2D_{i,j}D_{i,j+1}}{D_{i,j} + D_{i,j+1}}, \quad (34)$$

$$\text{Geometric: } D_{i,j+\frac{1}{2}} \approx \sqrt{D_{i,j}D_{i,j+1}}. \quad (35)$$

The harmonic mean is often preferred for diffusion coefficients to ensure monotonicity, but the code supports all three. Similarly for z-faces. These face-averaged coefficients $D_{i,j\pm 1/2}^n$ are then frozen at the old time t^n during the implicit solve.

4 Radial Diffusion Operator

4.1 Conservative Form

The radial part of the diffusion operator with space-dependent $D(r, z)$ is:

$$\mathcal{L}_r[E] = \frac{1}{r} \frac{d}{dr} \left(r D \frac{\partial E}{\partial r} \right). \quad (36)$$

This is the correct conservative form when D varies with position. We discretize by defining radial fluxes at cell faces:

$$F_{i,j+\frac{1}{2}}^r \equiv r_{j+\frac{1}{2}} D_{i,j+\frac{1}{2}}^n \frac{E_{i,j+1}^{n+1} - E_{i,j}^{n+1}}{\Delta r_{j+\frac{1}{2}}}, \quad \Delta r_{j+\frac{1}{2}} = r_{j+1} - r_j. \quad (37)$$

The divergence over a control volume at (i, j) is then:

$$\left[\frac{1}{r} \frac{d}{dr} \left(r D \frac{\partial E}{\partial r} \right) \right]_{i,j} \approx \frac{1}{r_j \Delta r_j} \left(F_{i,j+\frac{1}{2}}^r - F_{i,j-\frac{1}{2}}^r \right), \quad (38)$$

where Δr_j is the width of the control volume around node j (typically the average of $\Delta r_{j-\frac{1}{2}}$ and $\Delta r_{j+\frac{1}{2}}$).

4.2 Discrete Form for Interior Nodes

For interior nodes $1 \leq j \leq N_r - 2$, expanding the space-dependent operator:

$$\left[\frac{1}{r} \frac{d}{dr} \left(r D \frac{\partial E}{\partial r} \right) \right]_{i,j} \approx \frac{1}{r_j \Delta r_j} \left[r_{j+\frac{1}{2}} D_{i,j+\frac{1}{2}}^n \frac{E_{i,j+1}^{n+1} - E_{i,j}^{n+1}}{\Delta r_{j+\frac{1}{2}}} - r_{j-\frac{1}{2}} D_{i,j-\frac{1}{2}}^n \frac{E_{i,j}^{n+1} - E_{i,j-1}^{n+1}}{\Delta r_{j-\frac{1}{2}}} \right]. \quad (39)$$

Here, $D_{i,j+\frac{1}{2}}^n$ is the face-averaged diffusion coefficient at time t^n . Rearrange as a linear combination:

$$\approx \frac{r_{j+\frac{1}{2}} D_{i,j+\frac{1}{2}}^n}{r_j \Delta r_j \Delta r_{j+\frac{1}{2}}} E_{i,j+1}^{n+1} - \frac{r_{j+\frac{1}{2}} D_{i,j+\frac{1}{2}}^n}{r_j \Delta r_j \Delta r_{j+\frac{1}{2}}} E_{i,j}^{n+1} - \frac{r_{j-\frac{1}{2}} D_{i,j-\frac{1}{2}}^n}{r_j \Delta r_j \Delta r_{j-\frac{1}{2}}} E_{i,j}^{n+1} + \frac{r_{j-\frac{1}{2}} D_{i,j-\frac{1}{2}}^n}{r_j \Delta r_j \Delta r_{j-\frac{1}{2}}} E_{i,j-1}^{n+1}. \quad (40)$$

Define the effective cylindrical weights (with space-dependent coefficients):

$$\lambda_{i,j+\frac{1}{2}}^{r,n} \equiv \frac{\Delta t \cdot r_{j+\frac{1}{2}} D_{i,j+\frac{1}{2}}^n}{r_j \Delta r_j \Delta r_{j+\frac{1}{2}}}, \quad (41)$$

$$\lambda_{i,j-\frac{1}{2}}^{r,n} \equiv \frac{\Delta t \cdot r_{j-\frac{1}{2}} D_{i,j-\frac{1}{2}}^n}{r_j \Delta r_j \Delta r_{j-\frac{1}{2}}}. \quad (42)$$

Then the discrete radial contribution to the implicit step is:

$$\Delta t \left[\frac{1}{r} \frac{d}{dr} \left(r D^n \frac{\partial E}{\partial r} \right) \right]_{i,j}^{n+1} \approx \lambda_{i,j+\frac{1}{2}}^{r,n} E_{i,j+1}^{n+1} - \left(\lambda_{i,j+\frac{1}{2}}^{r,n} + \lambda_{i,j-\frac{1}{2}}^{r,n} \right) E_{i,j}^{n+1} + \lambda_{i,j-\frac{1}{2}}^{r,n} E_{i,j-1}^{n+1}. \quad (43)$$

5 Axial Diffusion Operator

5.1 Continuous Form

The axial part with space-dependent $D(r, z)$:

$$\mathcal{L}_z[E] = \frac{\partial}{\partial z} \left(D \frac{\partial E}{\partial z} \right). \quad (44)$$

5.2 Discrete Form for Interior Nodes

For interior axial nodes $i = 1, \dots, N_z - 2$, the space-dependent operator is approximated as:

$$\left[\frac{\partial}{\partial z} \left(D \frac{\partial E}{\partial z} \right) \right]_{i,j} \approx \frac{1}{(\Delta z)^2} [D_{i+1,j}^n E_{i+1,j}^{n+1} - (D_{i+1,j}^n + D_{i,j}^n) E_{i,j}^{n+1} + D_{i,j}^n E_{i-1,j}^{n+1}]. \quad (45)$$

However, for simplicity and to maintain stability, a common approximation averages the diffusion coefficient to the cell center:

$$D_{i,j}^n \approx \frac{D_{i+1,j}^n + D_{i,j}^n + D_{i-1,j}^n}{3}, \quad (46)$$

which then gives the simpler form:

$$\left[\frac{\partial}{\partial z} \left(D \frac{\partial E}{\partial z} \right) \right]_{i,j} \approx D_{i,j}^n \frac{E_{i+1,j}^{n+1} - 2E_{i,j}^{n+1} + E_{i-1,j}^{n+1}}{(\Delta z)^2}. \quad (47)$$

With implicit time stepping:

$$\Delta t \left[\frac{\partial}{\partial z} \left(D^n \frac{\partial E}{\partial z} \right) \right]_{i,j}^{n+1} \approx \frac{\Delta t \cdot D_{i,j}^n}{(\Delta z)^2} [E_{i+1,j}^{n+1} - 2E_{i,j}^{n+1} + E_{i-1,j}^{n+1}]. \quad (48)$$

Define the axial weight:

$$\lambda_{i,j}^{z,n} \equiv \frac{\Delta t \cdot D_{i,j}^n}{(\Delta z)^2}. \quad (49)$$

Then:

$$\Delta t \left[D \frac{\partial^2 E}{\partial z^2} \right]_{i,j}^{n+1} \approx \lambda_{i,j}^{z,n} E_{i+1,j}^{n+1} - 2\lambda_{i,j}^{z,n} E_{i,j}^{n+1} + \lambda_{i,j}^{z,n} E_{i-1,j}^{n+1}. \quad (50)$$

6 Complete Interior 5-Point Stencil

6.1 Combining All Terms

Start from Eq. (27) and apply time discretization:

$$\left(\frac{1}{\Delta t} + C^n\right) E_{i,j}^{n+1} - [\Delta t \mathcal{L}_r + \Delta t \mathcal{L}_z] E_{i,j}^{n+1} = \frac{E_{i,j}^n}{\Delta t} + C^n U_{i,j}^n. \quad (51)$$

Substitute the discrete forms from Eqs. (43) and (50):

$$\left(\frac{1}{\Delta t} + C^n\right) E_{i,j}^{n+1} - \left[\lambda_{i,j+\frac{1}{2}}^{r,n} E_{i,j+1}^{n+1} - (\lambda_{i,j+\frac{1}{2}}^{r,n} + \lambda_{i,j-\frac{1}{2}}^{r,n}) E_{i,j}^{n+1} + \lambda_{i,j-\frac{1}{2}}^{r,n} E_{i,j-1}^{n+1}\right] \quad (52)$$

$$- [\lambda_{i,j}^{z,n} E_{i+1,j}^{n+1} - 2\lambda_{i,j}^{z,n} E_{i,j}^{n+1} + \lambda_{i,j}^{z,n} E_{i-1,j}^{n+1}] = \frac{E_{i,j}^n}{\Delta t} + C^n U_{i,j}^n. \quad (53)$$

Collect all terms on the left:

$$\left(\frac{1}{\Delta t} + C^n + \lambda_{i,j+\frac{1}{2}}^{r,n} + \lambda_{i,j-\frac{1}{2}}^{r,n} + 2\lambda_{i,j}^{z,n}\right) E_{i,j}^{n+1} \quad (54)$$

$$- \lambda_{i,j+\frac{1}{2}}^{r,n} E_{i,j+1}^{n+1} - \lambda_{i,j-\frac{1}{2}}^{r,n} E_{i,j-1}^{n+1} - \lambda_{i,j}^{z,n} E_{i+1,j}^{n+1} - \lambda_{i,j}^{z,n} E_{i-1,j}^{n+1} \quad (55)$$

$$= \frac{E_{i,j}^n}{\Delta t} + C^n U_{i,j}^n. \quad (56)$$

Rearrange into standard form:

diagonal coeff: $\frac{1}{\Delta t} + C^n + \lambda_{i,j+\frac{1}{2}}^{r,n} + \lambda_{i,j-\frac{1}{2}}^{r,n} + 2\lambda_{i,j}^{z,n},$ r-neighbors: $-\lambda_{i,j\pm\frac{1}{2}}^{r,n},$ z-neighbors: $-\lambda_{i,j}^{z,n},$ RHS: $\frac{E_{i,j}^n}{\Delta t} + C^n U_{i,j}^n.$	(57)
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This is the full interior 5-point stencil (or 4-point in 1D) used in the implicit solver.

7 Boundary Conditions

7.1 Axial Boundary Conditions

7.1.1 Dirichlet Boundary at $z = 0$ (Drive)

At the irradiated surface, we impose:

$$E_{0,j}^{n+1} = E_{\text{drive}}(t^{n+1}, j), \quad (58)$$

where $E_{\text{drive}}(t)$ is computed from the imposed drive temperature $T_D(t)$ via $E_{\text{drive}} = aT_D^4$ with $a = 4\sigma_{\text{SB}}/c$.

This node is eliminated from the linear system; instead, its value is imposed after the solve.

7.1.2 Marshak Boundary at $z = 0$

Alternatively, the Marshak boundary condition relates the boundary value to the outward flux:

$$aT_D^4(t) = E(0, j, t) - \frac{2D(0, j, t)}{c} \left. \frac{\partial E}{\partial z} \right|_{z=0}. \quad (59)$$

Using backward differencing:

$$\left. \frac{\partial E}{\partial z} \right|_{z=0} \approx \frac{E_{1,j}^{n+1} - E_{0,j}^{n+1}}{\Delta z}. \quad (60)$$

Substituting:

$$aT_D^4(t^{n+1}) = E_{0,j}^{n+1} - \frac{2D_{0,j}^n}{c} \frac{E_{1,j}^{n+1} - E_{0,j}^{n+1}}{\Delta z}. \quad (61)$$

Multiply by Δz and rearrange:

$$aT_D^4(t^{n+1})\Delta z = E_{0,j}^{n+1}\Delta z - \frac{2D_{0,j}^n}{c} (E_{1,j}^{n+1} - E_{0,j}^{n+1}). \quad (62)$$

$$aT_D^4(t^{n+1})\Delta z = E_{0,j}^{n+1} \left(\Delta z + \frac{2D_{0,j}^n}{c} \right) - \frac{2D_{0,j}^n}{c} E_{1,j}^{n+1}. \quad (63)$$

Define $\alpha_j = \frac{2D_{0,j}^n}{c\Delta z}$. Then:

$$aT_D^4(t^{n+1}) = (1 + \alpha_j) E_{0,j}^{n+1} - \alpha_j E_{1,j}^{n+1}. \quad (64)$$

This is the discrete Marshak row at $z = 0$, replacing the interior 5-point stencil for that location.

7.1.3 Dirichlet Boundary at $z = L_z$ (Bath)

At the far end:

$$E_{N_z-1,j}^{n+1} = E_{\text{bath}} = a(300 \text{ K})^4. \quad (65)$$

7.2 Radial Boundary Conditions

7.2.1 Axis Symmetry at $r = 0$

By cylindrical symmetry:

$$\left. \frac{\partial E}{\partial r} \right|_{r=0} = 0. \quad (66)$$

This is enforced by setting $E_{i,0} = E_{i,1}$, which removes the $1/r$ singularity.

7.2.2 Dirichlet Boundary at $r = R_{\text{tot}}$ (Fixed Bath)

Impose:

$$E_{i,N_r-1}^{n+1} = E_{\text{bath}}. \quad (67)$$

7.2.3 Neumann Boundary at $r = R_{\text{tot}}$ (No Flux)

Impose:

$$\left. \frac{\partial E}{\partial r} \right|_{r=R_{\text{tot}}} = 0, \quad (68)$$

which eliminates the outer-face flux contribution in the stencil.

7.2.4 Marshak Boundary at $r = R_{\text{tot}}$ (Radiating Wall)

The Marshak condition at the outer surface is:

$$aT_{\text{wall}}^4(t) = E(N_r - 1, i, t) - \frac{2D(N_r - 1, i, t)}{c} \left. \frac{\partial E}{\partial r} \right|_{r=R_{\text{tot}}}. \quad (69)$$

Using backward differencing:

$$\left. \frac{\partial E}{\partial r} \right|_{r=R_{\text{tot}}} \approx \frac{E_{i,N_r-1}^{n+1} - E_{i,N_r-2}^{n+1}}{\Delta r_{\text{outer}}}. \quad (70)$$

Substituting and rearranging similarly to Eq. (64):

$$\boxed{aT_{\text{wall}}^4(t^{n+1}) = (1 + \alpha_i) E_{i,N_r-1}^{n+1} - \alpha_i E_{i,N_r-2}^{n+1}}, \quad (71)$$

where $\alpha_i = \frac{2D_{i,N_r-1}^n}{c \Delta r_{\text{outer}}}$.

This row is included in the linear system to replace the interior 5-point stencil at the outer radial boundary.

8 Linear System Assembly

8.1 Unknown Vector Organization

The implicit solve solves for E^{n+1} on an interior subset of the grid. Typically:

- If Marshak boundary at $z = 0$: unknowns are $E_{i,j}^{n+1}$ for $i = 0, \dots, N_z - 2$ and $j = 0, \dots, N_r - 1$.
- If Dirichlet at $z = 0$: unknowns are $E_{i,j}^{n+1}$ for $i = 1, \dots, N_z - 2$ and $j = 0, \dots, N_r - 1$.
- Boundary nodes at $z = L_z$ and possibly $r = R$ are either imposed after the solve or handled specially.

The total number of unknowns is:

$$n_{\text{unknown}} = n_z \times n_r, \quad (72)$$

where n_z and n_r are the counts of unknown rows in each direction.

8.2 Row-Wise Assembly

For each unknown (i, j) , the corresponding matrix row contains:

- Diagonal entry: coefficient of $E_{i,j}^{n+1}$,
- Off-diagonal entries: coefficients of neighboring $E_{i\pm 1,j}^{n+1}$ and $E_{i,j\pm 1}^{n+1}$,
- RHS: constant vector entries.

The sparsity pattern is structured: each row has at most 5 nonzeros (2D grid), or fewer at boundaries.

8.3 CSR Storage

To store the matrix efficiently in Compressed Sparse Row (CSR) format:

$$\text{indptr} : \text{row start pointers (length } n_{\text{unknown}} + 1), \quad (73)$$

$$\text{indices} : \text{column indices of stored values}, \quad (74)$$

$$\text{data} : \text{numeric coefficients}. \quad (75)$$

For each row, we store only the nonzero entries in sorted column order, reducing memory from $O(n_{\text{unknown}}^2)$ dense to $O(n_{\text{unknown}})$ sparse.

9 Material Energy Update

After solving for E^{n+1} , update U^{n+1} using Eq. (17):

$$U_{i,j}^{n+1} = \frac{U_{i,j}^n + A_{i,j}^n E_{i,j}^{n+1}}{1 + A_{i,j}^n}, \quad (76)$$

where $A_{i,j}^n = \beta_{i,j}^n \Delta t \cdot \chi \cdot c \sigma_{i,j}^n$.

This is a local (cell-by-cell) update, not requiring a linear solve.

10 Self-Similar Model Integration

In the self-similar model, the coupling parameter $\beta(T)$ (hence β^n at the old time) and the heat capacity are related to material parameters via:

$$\beta(T) = C_{vR}(T)/C_{vm}(T), \quad (77)$$

which is the ratio of radiation to material heat capacity (dimensionless). This is temperature-dependent and must be evaluated at each location from the old material energy U^n to get $\beta_{i,j}^n$.

The opacity and heat capacity are functions of temperature:

$$\sigma(T) = \frac{1}{gT^\alpha \rho^{-\lambda-1}}, \quad (78)$$

$$Cv_m(T) = f\beta_{\text{exp}} T^{\beta_{\text{exp}}-1} \rho^{-\mu+1}, \quad (79)$$

$$Cv_R(T) = 4aT^3. \quad (80)$$

At each timestep, $T^n = (U^n/a)^{1/4}$ is computed from the old material energy, and all temperature-dependent quantities (D^n, σ^n, C^n) are evaluated there.

A Quick Reference: Final Stencil Summary

A.1 Interior 5-Point Stencil

For interior nodes (i, j) with $1 \leq i \leq N_z - 2$ and $1 \leq j \leq N_r - 2$ (or adjusted ranges for Marshak):

A.1.1 Matrix Coefficients

$$a_{\text{self}} = \frac{1}{\Delta t} + C^n + \lambda_{i,j+\frac{1}{2}}^{r,n} + \lambda_{i,j-\frac{1}{2}}^{r,n} + 2\lambda_{i,j}^{z,n}, \quad (81)$$

$$a_{i+1,j} = -\lambda_{i,j}^{z,n}, \quad (82)$$

$$a_{i-1,j} = -\lambda_{i,j}^{z,n}, \quad (83)$$

$$a_{i,j+1} = -\lambda_{i,j+\frac{1}{2}}^{r,n}, \quad (84)$$

$$a_{i,j-1} = -\lambda_{i,j-\frac{1}{2}}^{r,n}. \quad (85)$$

A.1.2 RHS

$$b_{i,j} = \frac{E_{i,j}^n}{\Delta t} + C_{i,j}^n U_{i,j}^n, \quad (86)$$

where $C_{i,j}^n = \frac{c\sigma_{i,j}^n}{1+A_{i,j}^n}$ and $A_{i,j}^n = \beta_{i,j}^n \Delta t \cdot \chi \cdot c\sigma_{i,j}^n$.

A.2 Marshak Rows at $z = 0$

$$a_{\text{self}} = 1 + \alpha_j, \quad \alpha_j = \frac{2D_{0,j}^n}{c\Delta z}, \quad (87)$$

$$a_{1,j} = -\alpha_j, \quad (88)$$

$$b_{0,j} = aT_D^4(t^{n+1}). \quad (89)$$

A.3 Marshak Rows at $r = R_{\text{tot}}$

$$a_{\text{self}} = 1 + \alpha_i, \quad \alpha_i = \frac{2D_{i,N_r-1}^n}{c \Delta r_{\text{outer}}}, \quad (90)$$

$$a_{i,N_r-2} = -\alpha_i, \quad (91)$$

$$b_{i,N_r-1} = aT_{\text{wall}}^4(t^{n+1}). \quad (92)$$